

FORM 2

THE PATENTS ACT, 1970 (39 of 1970) & the Patents Rules 2003

COMPLETE SPECIFICATION (SECTION 10 and Rule 13)

1. Title of the Invention:

A method to generate images and a set of segmentation masks corresponding to the generated images and a system thereof.

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Complete Specification:

The following specification describes and ascertains the nature of this invention and the manner in which it is to be performed

Field of the invention

[001] The present invention relates to the field of generative artificial intelligence
5 and more specifically to a method to generate images and a set of segmentation masks corresponding to the generated images by an AI system.

[002] Background of the invention

10 [003] Autonomous driving systems now deploy perception models for object recognition. Training these models requires a large amount of diverse imagery containing relevant traffic objects and their annotations. However, any data collection effort to achieve this will fall short, given the enormity of the effort.

15 [004] According to prior art, the usual approach to solving this problem is to make use of open-source Generative Models which are based on Vision-Language Frontier Maps (VLFMs). However, one of the fundamental challenges of this approach is that specific objects are not well represented in the underlying foundation models. For example, a speed bump or a traffic specific object such as
20 a road sign or road marking may not be well represented since the FMs are usually trained using commonly seen world objects.

[005] Other works propose the use of NERFs (Neural Radiance Fields) and Gaussian Splatting and Gaussian splatting to re-create the 3D scene over time and then to
25 manually insert objects into the 3D space. Such an approach works only when there are static scenes and significant interventions are required to compensate for moving objects. Furthermore, the insertion of objects into the 3D scene requires significant human effort. This work focuses on creating a system capable of generating exhaustive dataset of photo-realistic image and segmentation mask pairs given only

a few images of the object. These pairs can then be used to train a segmentation model for segmentation of the new object.

5 [006] The present disclosure proposes to fine tune pre-trained diffusion models along with swappable adapters for realistic data generation with high fidelity. The variations in the images will be guided by different text prompts and the user can variations such as changing the object location, changing object properties such as color or shape, viewpoint, and other semantic modifications. Additionally, the present disclosure proposes for localization of the said objects
10 in the generated image in the form of approximate bounding boxes or semantic segmentation masks. The generated images are fed into a filtering module that checks the image for plausibility (realistic representation). If plausible, the images together with the bounding boxes are fed into a segmentation model to refine and establish real ground truth segmentations, facilitating enhanced model
15 training.

[007] Upon implementation, the present disclosure proposes a resource efficient system which can be used to generate a diverse set of images of a new object given only a handful of images. Further, implementation of the present disclosure
20 leads to generation of precise ground truths (bounding boxes and segmentation masks) for the generated images. The present disclosure further proposes a filtering module to reject implausible scenes. Consequently, the generated images and the bounding boxes can be used to train a segmentation model.

25 [008] The present disclosure advantageously provides for creation of huge datasets for out-of-distribution (OOD) classes along with their ground truths, which then can be used to segment the O.O.D object. The same further boosts the under-

represented classes by generating new and diverse images and masks, thus improving the prediction accuracy of segmentation models.

5 [009] Further, compared to other approaches in the prior arts that are based on physics based models or NERFs, the present approach requires significantly less effort and can be easily incorporated in existing machine learning operations pipelines for perception algorithm training.

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Brief description of the accompanying drawings

[0010] An embodiment of the invention is described with reference to the following accompanying drawings:

15 [0011] Figure 1 depicts an AI system (10) to generate images and a set of segmentation masks (16) corresponding to the generated images (9), according to an embodiment of the present disclosure.

20 [0012] Figure 2 depicts a flow chart for a method to generate images and a set of segmentation masks (16) corresponding to the generated images (9), according to an embodiment of the present disclosure.

[0013] Figure 3 illustrates the method of generating bounding boxes (8) by the bounding box module (15), according to an embodiment of the present disclosure.

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[0014] Figure 4 illustrates the method of filtering the generated images (9) by the filtering module (2), according to an embodiment of the present disclosure.

Detailed description of the drawings:

5 [0015] The present invention will now be described by way of example, with reference to accompanying drawings. Throughout all the figures, same or corresponding elements may generally be indicated by same reference numerals. These depicted embodiments are to be understood as illustrative of the invention and not as limiting in any way. It should also be understood that the figures are not necessarily to scale and that the embodiments are sometimes illustrated by
10 graphic symbols, phantom lines, diagrammatic representations, and fragmentary views. In predetermined instances, details which are not necessary for an understanding of the present invention, or which render other details difficult to perceive may have been omitted.

15 [0016] Referring to Figure 1, the same discloses an AI system (10) to generate images and a set of segmentation masks (16) corresponding to the generated images (9). Said AI system (10) comprises a Latent diffusion model (LDM (10)), a bounding-box module (15) (see figure 3) comprising an attention store(11), a filtering module (2), and a segmentation model (3).

20 [0017] Said LDM (10) is trained on a set of training images (6) and at least one training text prompt (7) corresponding to said set of images. Further, said LDM (10) is fine-tuned by a first Low rank adaptation module (4) (LoRa). Said LDM (10) is configured to receive at least one text prompt (7a) (see Figure 3) and
25 gaussian noise (20) as input to generate images (referred to as generated images (9)). Said LDM (10) is configured to extract a set of cross-attention blocks, said set of cross attention blocks being a cross-between attention features corresponding to the generated images (9), and the attention features

corresponding to the at least one text prompt (7a). The attention store (11) (See figure 3) is configured to store a subset of cross attention blocks in the attention store. Said subset of cross-attention blocks comprises low-rank features corresponding to the generated images (9) and the at least one text prompt (7a).

5 The bounding-box module (15) is adapted to generate bounding boxes (8) around at least one object in the generated images (9) based on the subsets of cross attention blocks. The filtering module (2) is configured to filter the generated images (9) and the bounding boxes (8) corresponding to said generated images (9) based on a comparison between the set of training images (6) and the

10 generated images (9). The filtering module outputs the filtered images (10). The segmentation model (3) (SM) is trained on the filtered images (10) and the bounding boxes (8) corresponding to said filtered images (10) to generate segmentation masks (16). Further, said segmentation module is fine-tuned by a second low rank adapter (5).

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[0018] The AI system (10) is further described in detail. It is important to understand some aspects of artificial intelligence (AI) technology and artificial intelligence (AI) based systems or artificial intelligence (AI) system. A person skilled in the art is expected to know the function of a Latent Diffusion Model which is a type

20 of deep generative model used for image generation, combining the strengths of diffusion models and latent space learning to efficiently create high-quality images. The present disclosure also covers the use of segmentation models. Segmentation models are used to classify pixels in an image, effectively dividing it into meaningful regions, they use encoder-decoder structures to output a

25 segmentation mask. The present disclosure further assumes that a person skilled in the art understands U-Net as an image autoencoder that combines convolutional neural networks (CNNs) with an encoder-decoder structure where the encoder extracts features and reduces dimensions through convolution and

the decoder up samples the feature maps using transposed convolution. The present disclosure further assumes that a person skilled in the art understands that the diffusion models generate images by gradually adding noise to an image and then learning to reverse this process where a model (like U-Net) is trained to remove noise, allowing it to generate new images from pure random noise. Further, the phrases ‘training’, ‘fine-tuning’ are limited to the meaning of training and AI model and fine tuning the AI model as known in the parlance of the field of AI technologies. AI system (10) is trained using the training data, the AI systems use the models to analyze the real time data and generate appropriate result. The models may be fine-tuned in real-time based on the results. It is further to be understood that the term ‘module’ used according to the present disclosure is a self-contained hardware or software component that interacts with the larger system. For example, a filter module as mentioned herein after can be a software residing in the system or a cloud or embodied within an electronic chip. Such chips are specialized silicon chips, which incorporate AI technology and are used for machine learning. The other components of the present disclosure will be clear with further reading of the present disclosure.

[0019] The AI system (10), according to the present disclosure is implemented in a computing device with specialized components optimized for parallel processing, memory access, and efficient deep learning computations. The computing device may comprise a General-purpose processor handling control logic, pre/post-processing, and scheduling AI tasks and adapted for data loading, model orchestration, and running lightweight AI models. The computing device may further comprise a Graphics Processing Unit (GPU) optimized for parallel processing and matrix operations in deep learning and for training & inference in AI models like transformers, diffusion models, and segmentation networks. The

computing device may further comprise Tensor Processing Unit (TPU), for deep learning workloads. The computing device may further comprise a high-Bandwidth Memory (HBM) essential for large AI models like Latent Diffusion models & Transformers and for fast data access for matrix multiplications & attention mechanisms. The computing device may further comprise matrix multiplication cores (Tensor Cores, AMX, XMX, etc.) for AI matrix operations (e.g., attention layers, convolutions, diffusion steps). Further, the computing device may have further essential components such as interconnects, network buses and the like. All the inputs received to the AI system (10) and outputs generated from the AI system (10) are through the corresponding input/output interfaces of the computing device. It is to be understood that the AI system (10) as described according to the present disclosure is an aggregation of AI models and filters combined to implement the method to generate images and a set of segmentation masks (16) corresponding to the generated images (9). According to an example, these components are executed as program codes in known frameworks such as pyTorch, tensor flow, hugging Face Diffusers, labml.ai and the like.

[0020] The components of the AI system (10) are further described. The AI system (10) comprises the Latent diffusion model (LDM(10)). Typically, a Latent Diffusion Model as known in the art first compresses images into a lower-dimensional latent space using an autoencoder (like a Variational Autoencoder (VAE)). The model gradually adds noise to this latent representation over multiple steps, making it look like pure noise. Then, a U-Net-based denoiser learns to reverse this noise process, reconstructing the original (or newly generated) image. LDMs are guided by text prompts (text-to-image), class labels, or other conditions (like depth maps, sketches, or segmentation masks). After the latent space is denoised, the autoencoder decodes it back into an image,

producing the final output. According to the present disclosure, the LDM (10) is trained on a set of training images (6) and at least one training text prompt (7) corresponding to said set of images. Further, said LDM(1) is fine tuned by a first Low rank adaptation module (LoRa).

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[0021] The first Low rank adaptation module, according to the present disclosure, is an adapter. Adapters are lightweight, trainable modules added to a pre-trained model to adapt it to new tasks or domains without modifying the original model's weights. They are particularly useful in transfer learning, where one wishes to fine-tune a large model efficiently without retraining all of its parameters. Instead of updating all model parameters, adapters introduce a small set of new trainable parameters, reducing computational cost and memory requirements. In other words, adapters are small feed-forward networks (often bottleneck layers) inserted within the transformer layers of a pre-trained model. During fine-tuning, only the adapter layers are trained while keeping the original model frozen. According to the present disclosure, the first low rank adapter is a Low-Rank Adaptation (LoRA) module.

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[0022] According to the present disclosure, referring to figure 3, said LDM(1) is configured to receive at least one text prompt (7a) and gaussian noise (20) as input to generate images. Said LDM (10) is configured to extract a set of cross-attention blocks, said set of cross attention blocks being a cross-between attention features corresponding to the generated images (9), and attention features corresponding to the at least one text prompt. In diffusion models, cross-attention blocks enable the model to interact with and incorporate external information (text prompts) during image generation, allowing for semantic consistency and control over the generated Image. Cross-attention blocks are a fundamental mechanism in diffusion models that enable interaction between

different types of input data, such as text-to-image (Stable Diffusion). According to an example of the present disclosure, the text prompt is encoded using a CLIP- (Contrastive Language-Image Pre-Training) based text encoder, and fed to the cross-attention blocks of the model. A Gaussian noise is provided as the input to the model which gets denoised iteratively over T timesteps.

[0023] According to the present disclosure, referring to figure 3, the attention store (11) is configured to store a subset of cross attention blocks in the attention store. Said subset of cross-attention blocks comprises low-rank features corresponding to the generated images (9) and the at least one text prompt (7a). According to an example, during image generation, attention maps from the U-Net's cross-attention layers are extracted. According to an example, if the cross-attention blocks have resolutions 96x96, 48x48, 24x24 in the U-Net's cross attention layers, the lower-level features from 24x24 are stored in the attention store. It is to be understood that the attention store (11) is termed to denote the cross-attention blocks stored within the U-Net architecture.

[0024] According to the present disclosure, the bounding-box module (15) comprising the attention store (11) is adapted to generate bounding boxes (8) around at least one object in the generated images(9) based on the subsets of cross attention blocks. Bounding boxes (8) can be extracted from cross-attention maps in latent diffusion models (LDMs(10)). From the subsets of cross-attention blocks extracted during image generation, cross attention maps can be obtained that visualize attention score computed inside the cross-attention blocks. According to an example, the attention map for a word like "traffic_signal" will highlight image regions related to that object. Attention maps across different timesteps (diffusion steps) are accumulated to get a stable heatmap. These

attention maps are then normalized using the SoftMax function to create a probabilistic distribution map. This distribution map serves as the basis for the subsequent task of selecting coordinates for bounding box. The magnitude of each point within this attention-derived distribution map is leveraged as a weight parameter to guide the selection process. This ensures that the selection of positions is influenced by the saliency and significance of the underlying attention-based features.

[0025] According to the present disclosure, referring to figure 4, the filtering module (2) is configured to filter the generated images (9) and the bounding boxes (8) corresponding to said generated images (9) based on a comparison between the set of training images (6) and the generated images (9). The generated images (9) are fed to the filtering module (2). The filtering module (2) extracts the features of the set of training images (6) and the generated images (9). The filtering module (2) employs a pre-trained encoder (14) to extract image features which represent the overall semantic distribution of the images. The extracted features of the generated images (9) are compared with the features of the set of training images (6). The generated images (9) are selected (filtered) only when the features of the generated images (9) semantically align with those of the set of training images (6).

[0026] According to the present disclosure, referring to figure 1, the segmentation model (3) is trained on the filtered images (10) and the bounding boxes (8) corresponding to said filtered images (10) to generate segmentation masks (16). Further, said segmentation module is fine-tuned by a second low rank adapter (5).

[0027] The method to generate images and a set of segmentation masks (16) corresponding to the generated images (9) is further described, Referring to Figure 2, the same depicts a flowchart for a method (100) to generate images and a set of segmentation masks (16) corresponding to the generated images (9) by an AI system (10). It is to be understood the method (100) as described hereafter is implemented by the AI system (10) as described in Figure 1. Said AI system (10) comprises a Latent diffusion model (LDM(10)) , a bounding-box module (15) comprising an attention store (11), a filtering module (2) and a segmentation model(3).

[0028] According to the present disclosure, referring to figure 2, the method has several steps from step 101 to step 107 and sub-steps (107a), (107b), (107c), (107d) to the step (107). The steps are listed herein: Step (101) is training the LDM (10) by inputting a set of training images (6) and at least one training text prompt (7) corresponding to said set of images to the LDM(10). Step (102) is fine tuning said LDM (10) by means of a first low-rank adapter (4). Step (103) is inputting the fine-tuned LDM(1) with at least one text prompt and gaussian noise (20) to generate images. Step (104) is extracting a set of cross-attention blocks, said set of cross attention blocks being a cross between: attention features corresponding to the generated images (9), and attention features corresponding to the at least one text prompt (7a). Step (105) is storing a subset of cross attention blocks in the attention store, said subset of cross-attention blocks comprising low-rank features corresponding to the generated images (9) and the at least one text prompt. Step (106) is generating bounding boxes (8) by the bounding box module (15) around at least one object in the generated images (9) based on the subset of cross attention blocks. Step (107) is filtering the generated images (9) and the bounding boxes(8) corresponding to said generated images(9) by a filtering module (2) based on a comparison between the set of training

images (6) and the generated images(9). Step (108) is training the segmentation model (3) based on the filtered images (10) and the bounding boxes (8) corresponding to the filtered images (10) to generate the segmentation masks (16). Step (109) is fine tuning the said segmentation module by a second low-rank adapter (5). The step (107) further comprises the step (107a) of generating a first semantic distribution (13) of the set of training images (6) by the encoder(14), step (107b) of generating a second semantic distribution (12) of the generated images (9) by the encoder(14), step (107c) of computing a similarity score threshold between the first and second semantic distribution (12) and step (107d) of filtering images based on said similarity score threshold.

[0029] The steps are further described in detail. According to the present disclosure, the Step (101) is training the LDM (1) by inputting the set of training images (6) and the at least one training text prompt (7) corresponding to said set of images to the LDM(10). The Step (102) is fine tuning said LDM(1) by means of a first low-rank adapter (4).

[0030] According to an example, given the pre-trained text-to-image latent diffusion model, $\mathcal{M}$ and the set of images of the object to learn, $\mathcal{J}$, the LDM $\mathcal{M}$ is fine tuned using a fine-tuning technique known in the art as DreamBooth, $\mathcal{D}$. The training criteria is to evaluate the alignment between input training images, $\mathcal{J}$ and the generated images (9). To measure the alignment of a given training image, I_{input} and the generated image, I_{gen} the distance is obtained as $\|I_{input} - I_{gen}\|$. To reduce the training parameters, in essence, to reduce the resources and time required for fine-tuning, a low-rank matrix factorization technique, herein called as Low Rank Adapters is used which reduces the parameters required to train the model. According to an example, in order to fine-tune the LDM(10), a generic prompt is created which summarizes the overall set of images of at least one

object in the form of "<Identifier> <Object Name(noun)> Description". This prompt is encoded using a CLIP-based text encoder(14), $\mathcal{T}$ and fed to the cross-attention blocks of the LDM(10), along with a Gaussian noise (20). The LDM(1) then generates images. A pixel-wise distance between the generated images (9) and the training images is calculated and is used to update the cross-attention blocks in order to teach model the specifications of the at least one object. During fine tuning, instead of optimizing all the features, only the low-rank features are optimized. According to an embodiment of the present disclosure, the trained LDM(1) can be shared and used to build many small LoRA modules for different objects. The shared LDM(1) can be freezed while the features of objects can be switched by replacing the LoRA module. This reduces the storage requirement.

[0031] Step (103) is inputting the fine-tuned LDM (10) with at least one text prompt and gaussian noise (20) to generate images. According to an example, the fine tuned LDM (1) is used to generate diverse object scenes using different text prompts. The text prompt is encoded using a CLIP-based text encoder (14), $\mathcal{T}$ and fed to the cross-attention blocks of the LDM(10). A Gaussian noise (20) is provided as the input to the model which gets denoised iteratively over T timesteps. The text prompt guides the location, style and shape of the object to be rendered in the scene.

[0032] Referring to Figure 3, step (104) is extracting a set of cross-attention blocks, said set of cross attention blocks being a cross between a set attention features corresponding to the generated images (9), and the attention features corresponding to the at least one text prompt (7a). Step (105) is storing a subset of cross attention blocks in the attention store (11), said subset of cross-attention blocks comprising low-rank features corresponding to the generated images (9) and the at least one text prompt. According to an example, during image

generation, attention maps from the U-Net's cross-attention layers are extracted. According to an example, if the cross-attention blocks have resolutions 96x96, 48x48, 24x24 in the U-Net's cross attention layers, the lower-level features from 24x24 are stored in the attention store. Consequently, Step (106) is generating
5 bounding boxes (8) by the bounding box module (15) around at least one object in the generated images(9) based on the subsets of cross attention blocks.

[0033] Referring to Figure 4, the step (107) is filtering the generated images (9) and the bounding boxes (8) corresponding to said generated images (9) by a filtering
10 module (2) based on a comparison between the set of training images (6) and the generated images (9). Said filtering module (2) comprises an encoder(14). Step (107) has sub-steps (107a), (107b), (107c) and (107d). Step (107a) is generating a first semantic distribution (13) of the set of training images (6) by the encoder (14). Step (107b) is generating a second semantic distribution (12) of the
15 generated images (9) by the encoder (14). Step (107c) is computing a similarity score threshold between the first and second semantic distribution (12) and step (107d) is filtering images based on said similarity score threshold. According to an example, the generated images (9) are fed to the filtering module (2). The filtering module (2) extracts the features of the set of training images (6) and the
20 generated images (9). The filtering module (2) employs a pre-trained encoder (14) to extract image features which represent the overall semantic distribution of the images. The extracted features of the generated images (9) are compared with the features of the set of training images (6). The generated images (9) are selected (filtered) only when the features of the generated images(9) semantically
25 align with those of the set of training images (6). A similarity score is computed between the features of the set of training images (6) and generated images (9) and if the said similarity score is above a threshold, the filter allows the image to pass through it.

[0034] According to the present disclosure, referring to figure 1 and 2, Step (108) is training the segmentation model (3) based on the filtered images (10) and the bounding boxes(8) corresponding to the filtered images (10) to generate the segmentation masks (16). Step (109) is fine tuning the said segmentation module by a second low-rank adapter (5). According to the present disclosure, the bounding boxes(8) and the generated images(9) are used to generate the segmentation masks (16). These (bounding boxes (8) and generated images(9)) are fed as an input to a class-agnostic semantic segmentation model(3) that generates the segmentation masks (16) corresponding to the object over which the said segmentation model(3) is trained. The segmentation model(3) is a foundational model trained on diverse data and can segment new objects when trained with a prior object. To train the segmentation model(3) to segment the new object, the pair of generated images(9) (filtered images) and the segmentation masks (16) generated in Step (108) are used. Furthermore, to facilitate parameter efficient training of the segmentation model (3) a second LoRA (5) is deployed, which reduces parameter by a significant margin. The trained model can then be used to segment a new object in any new image.

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[0035] According to the present disclosure, said LDM(1) is fine tuned with a first set of Low-rank adapters (4a, 4b, 4c, 4d), wherein every LoRA in the first set of LoRa is tuned for an object selected from a set of objects in the set of training images (6). The first low rank adapter is selected from said first set of Low-rank adapters. According to the present disclosure, said segmentation model (3) is fine tuned by a second set of Low-rank adapters (5a, 5b, 5c, 5d), wherein every LoRA in the second set of LoRa is tuned for an object selected from a set of objects in the generated images(9). In other words, the low rank adapters in the

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first set of low rank adapters are swappable. Similarly, the low rank adapters in the second set of low rank adapters are swappable. According to an example, if the an adapter functions better to segment traffic light (in a segmentation model(3)) or to generate traffic lights (in an LDM(10)) and the other functions better to segment pedestrians or to generate pedestrians, the same can be swapped with each other according to the desired functionality.

[0036] It must be understood that the embodiments explained in the above detailed description are only illustrative and do not limit the scope of this invention. Any modification to the method and the system described above are envisaged and form a part of this invention. The scope of this invention is limited only by the claims.

We Claim:

1. A method to generate images and a set of segmentation masks (16) corresponding to the generated images (9) by an AI system (10), said AI system (10) comprising a Latent diffusion model (LDM (10)), a bounding-box module (15) comprising an attention store (11), a filtering module (2) and a segmentation model (3), the method comprising the steps of:

- Training the LDM (10) by inputting a set of training images (6) and at least one training text prompt (7) corresponding to said set of images to the LDM (10),
- fine tuning said LDM (10) by means of a first low-rank adapter (4),
- inputting the fine-tuned LDM (10) with at least one text prompt (7a) and gaussian noise (20) to generate images,

-extracting a set of cross-attention blocks , said set of cross attention blocks being a cross between:

- attention features corresponding to the generated images (9), and

-attention features corresponding to the at least one text prompt(7a),

5 -storing a subset of cross attention blocks in the attention store (11), said subset of cross-attention blocks comprising low-rank features corresponding to the generated images (9) and the at least one text prompt,

10 -generating bounding boxes (8) by the bounding box module (15), said bounding boxes (8) generated around at least one object in the generated images (9) based on the subsets of cross attention blocks;

-filtering the generated images (9) and the bounding boxes (8) corresponding to said generated images(9) by a filtering module (2) based on a comparison between the set of training images (6) and the generated images(9),

15 -training the segmentation model (3) based on the filtered images (10) and the bounding boxes (8) corresponding to the filtered images (10) to generate the segmentation masks (16); and

-fine tuning the said segmentation module by a second low-rank adapter (5) .

20 2. The method as claimed in claim 1, wherein, filtering the generated images (9) by the filtering module (2), said filtering module comprising an encoder (14), the method comprising the steps of:

-generating a first semantic distribution (13) of the set of training images (6) by the encoder(14),

-generating a second semantic distribution (12) of the generated images(9) by the encoder(14),

5 -computing a similarity score threshold between the first and second set of semantic distribution,

-filtering images based on said similarity score threshold.

3. The method as claimed in Claim1, wherein, fine tuning said LDM (10)
10 comprises:

-fine tuning said LDM (10) with a first set of Low-rank adapters, wherein every LoRA in the first set of LoRa is tuned for an object selected from a set of objects, the training images comprising said set of objects.

4. The method as claimed in Claim 3, wherein, the first low rank adapter is
15 selected from said first set of Low-rank adapters.

5. The method as claimed in Claim 1, wherein, fine tuning said segmentation model (3) by a second low-rank adapter (5) comprises:

- fine tuning said segmentation with a second set of Low-rank adapters, wherein every LoRA in the second set of LoRa is tuned for an object selected from a set of
20 objects, the generated images(9) comprising said set of objects.

6. The method as claimed in Claim 5, wherein, the second low rank adapter is selected from said second set of Low-rank adapters.

7. An AI system (10) to generate images and a set of segmentation masks (16) corresponding to the generated images (9), said AI system (10) comprising:

- a Latent diffusion model (LDM (10)) , said LDM(10) trained on a set of training images (6) and at least one training text prompt (7) corresponding to said set of training images and said LDM(10) fine-tuned by a first LoRA, said LDM(10) configured to:

-receive at least one text prompt(7a) and gaussian noise (20) as input to generate images,

- extract a set of cross-attention blocks, said set of cross attention blocks being a cross between:

- attention features corresponding to the generated images (9), and

- attention features corresponding to the at least one text prompt;

-a bounding box module comprising an attention store (11) , said attention store configured to store a subset of cross attention blocks in the attention store, said subset of cross-attention blocks comprising low-rank features corresponding to the generated images(9) and the at least one text prompt;

-said bounding-box module adapted to generate bounding boxes(8) around at least one object in the generated images(9) based on the subsets of cross attention blocks;

-a filtering module (2) configured to filter the generated images (9) and the bounding boxes (8) corresponding to said generated images (9) based on a comparison between the set of training images (6) and the generated images(9); and

-a segmentation model (3) trained on the filtered images (10) and the bounding boxes (8) corresponding to said filtered images (10) to generate segmentation masks (16) and said segmentation module fine-tuned by a second low rank adapter module.

5 8. The AI system (10) as claimed in Claim 7, wherein, said LDM (10) is fine-tuned with a first set of Low-rank adapters, wherein every LoRA in the first set of LoRa is tuned for an object selected from a set of objects in the set of training images (6).

9. The method as claimed in Claim 8, wherein, the first low rank adapter is
10 selected from said first set of Low-rank adapters.

10. The method as claimed in Claim 7, wherein, said segmentation model (3) is fine-tuned by a second set of Low-rank adapters, wherein every LoRA in the second set of LoRa is tuned for an object selected from a set of objects in the generated images (9).

15 11. The method as claimed in Claim 8, wherein, the second low rank adapter is selected from said second set of Low-rank adapters.

Dated this 31st day of March , 2025

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(Digitally signed)
Siddharth Karkhanis
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On behalf of the applicants

5 **A method to generate images and a set of segmentation masks
corresponding to the generated images and a system thereof.**

Abstract

Disclosed are techniques to generate images and a set of segmentation masks (16)
10 corresponding to the generated images. Techniques include training an LDM and
fine tuning said LDM (10) by means of a first low-rank adapter (4) to generate
images (9), extracting a set of cross-attention blocks and storing a subset of cross
attention blocks in an attention store (11) and generating bounding boxes (8) based
on these subsets. Techniques further include filtering the generated images (9) and
15 the bounding boxes (8) corresponding to said generated images(9) by a filtering
module (2) and training the segmentation model (3) and fine tuning said
segmentation module by a second low-rank adapter (5). The low rank adapters are
swappable.

20 Figure 1

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